Minimum spanning trees: exercises

23-E01 Respecting cuts and quantifiers for ties

The five-vertex graph has ab4,ac2,bc1,bd5,cd3,ce6,de2,be4. Let A={ac,bc,de},S={a,b,c}. List all crossing edges, prove the cut respects A, and find a safe edge. For a triangle with all weights 1, assess whether any light edge belongs to some MST, every light edge belongs to every MST, and all light edges can be added together. Give concrete trees for each false statement. Finally prove that a uniquely heaviest cycle edge belongs to no MST.

23-E02 All Kruskal acceptances and rejections

Use the graph from 23-E01, with edge IDs0..7 in order ab,ac,bc,bd,cd,ce,de,be. Sort by (weight,ID). List the endpoint components before every scan, acceptance or rejection, and cumulative cost. When may the algorithm first stop? Lower be to 1 and recompute accepted order and cost. Does tie order affect correctness?

23-E03 Costs under different graph models

Compare a simple graph with n=1000,m=2000 and a multigraph with n=1000,m=10^7. State Kruskal's sorting and DSU terms and the costs of three Prim variants. Where is replacing log m by log n unjustified? If all weights are integers 0..9, what remains after replacing sorting? What is missing from an optimality argument based only on not creating a cycle?

23-E04 Competing Prim candidates

Run Prim from a on 23-E01. For every iteration give S, every unselected key and parent, selected edges, and cost. List all parent changes for b and e and adjacent extractions with decreasing keys. Why does updating with key[u]+w(u,v) solve a different problem? Give both outcomes on ab4,ac5,bc2.

23-E05 Certificates, uniqueness, and weight decreases

For T={bc1,ac2,de2,cd3}, list the T path, maximum path weight, and slack w(e)−M_T of every nontree edge, and prove uniqueness. Change be to 3 and then to 1. Give an optimal tree, its cost, and uniqueness in each case. Prove the one-edge decrease formula with a complete lower bound, rather than merely invoking the cycle property.

Minimum spanning trees: exercises (continued)

23-E06 Negative weights, parallel edges, and isolation

Vertices are {a,b,c,d}; edges are e0=aa:−100,e1=ab:−2,e2=ab:1,e3=bc:0,e4=ac:4, with isolated d. Run Kruskal. Is its result an MST or an MSF? When should Prim from a reject an input claimed to be connected? In a forest version, how is d included? Why must neither the negative self-loop nor∞ enter the cost?

23-E07 Deriving the cut certificate in reverse

For a candidate spanning tree T, prove that w(e)≥M_T for every non-self-loop nontree edge is equivalent to optimality, and strictness everywhere is equivalent to uniqueness. Test T′={ab1,ac2} in ab1,bc1,ac2 and give an improving exchange. Give a three-vertex graph with repeated input weights and a unique MST, refuting the necessity of distinct weights.

23-E08 Equal-cost exchange and the second edge

Let a graph have ab:1, cd:1, ad:2, bc:2, ac:4, bd:5. Set A={ab,cd}, S={a,b}, and T={ab,ad,cd}. List all crossing edges, prove T optimal, then exchange e=bc into T. Identify the cycle, removed edge, preservation of A, and both costs. Must every MST contain bc? Prove the second-lightest-edge claim for connected simple graphs, explain ties, and give a parallel-edge counterexample.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study